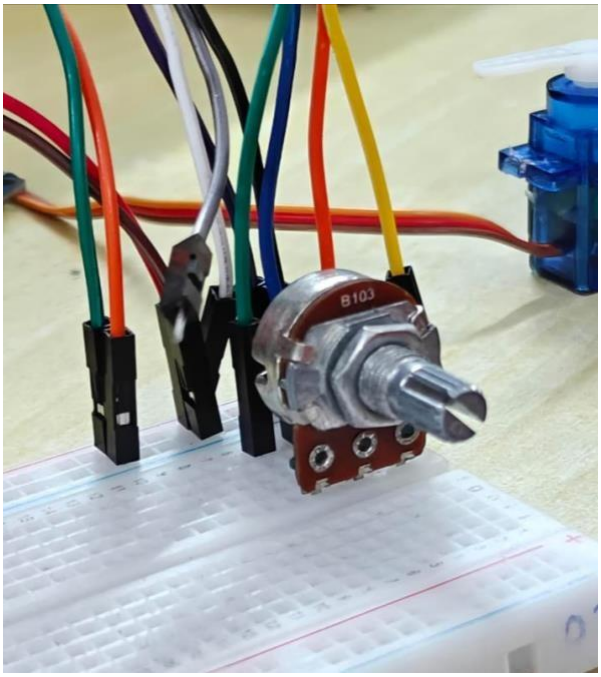
EXPERIMENT NO: 2

AIM: Interfacing and Controlling a Servo Motor using a Potentiometer (variable resistor)

ACTUATOR NAME: SG-90 :: Micro Servo Motor :: Rotation: 0° – 180° (Weight: 9g)

INPUT DEVICE: 10kΩ Potentiometer :: Type: Linear Variable Resistor



CONNECTION OF ARDUINO MEGA TO COMPONENTS:

- For the Micro Servo: The Red wire is VCC (5V), Brown is GND, and Orange is Signal (Pin 8).

- For the Potentiometer: The Left pin is GND, the Middle (Wiper) is Data (Pin A0), and the Right pin is VCC (5V).

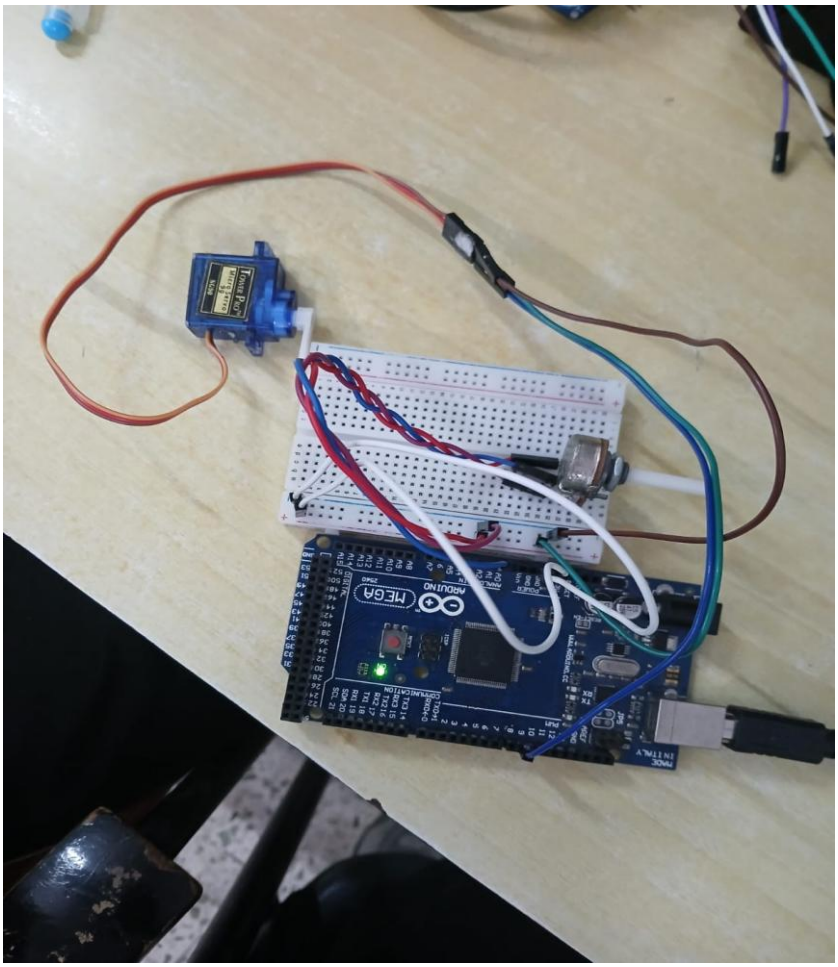
Code:

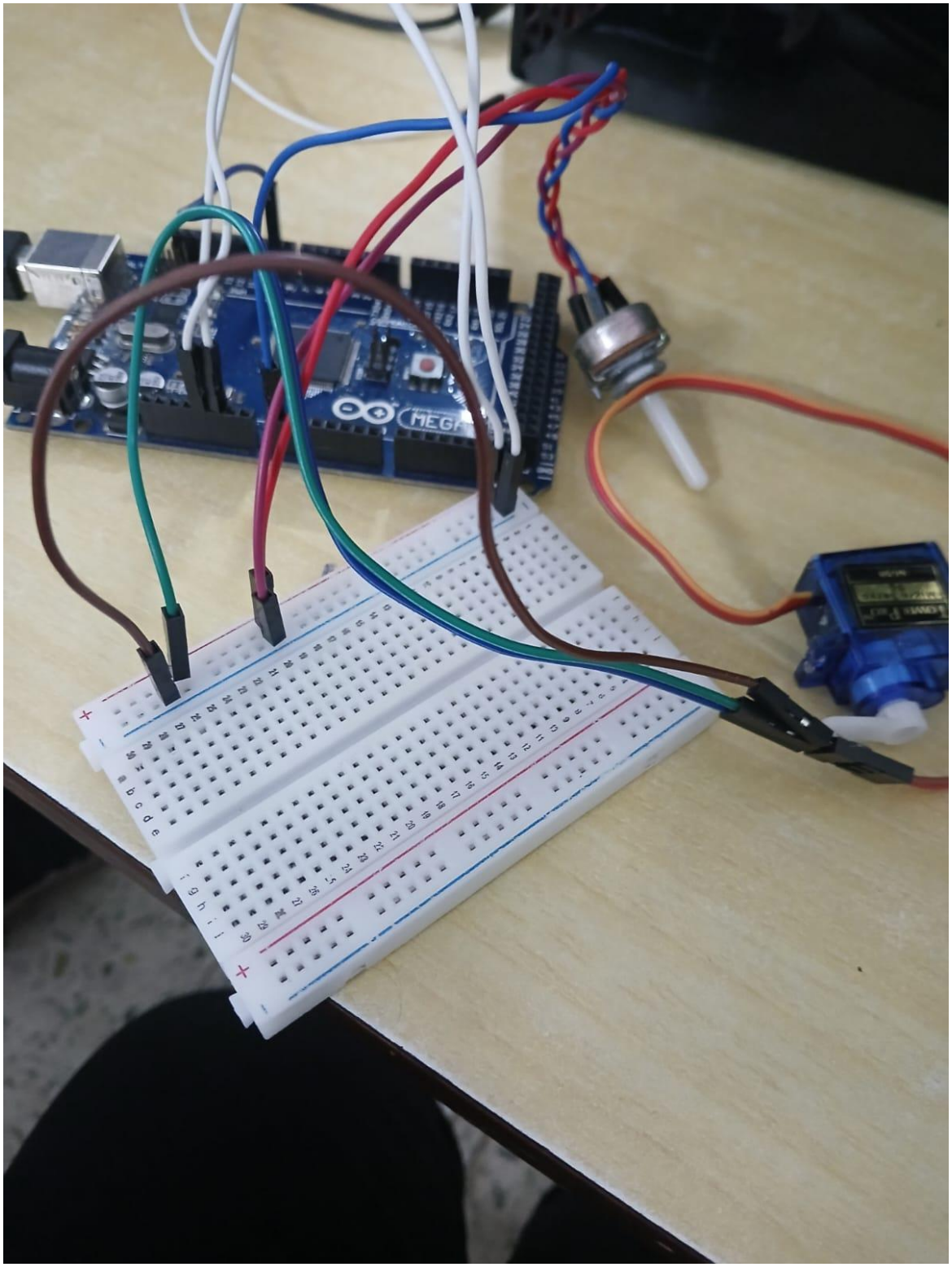
```
#include <Servo.h>
Servo myservo; // create servo object to control a servo int
servoPin = 8; // digital pin used to connect the servo int potpin =
A0; // analog pin used to connect the potentiometer int val; //
variable to read the value from the analog pin

void setup() {
myservo.attach(servoPin); // attaches the servo on pin 8 to the servo object
}

void loop() {
val = analogRead(potpin); // reads the value of the potentiometer (value between 0 and 1023)
val = map(val, 0, 1023, 0, 180); // scale it to use it with the servo (value between 0 and 180)
myservo.write(val); // sets the servo position according to the scaled value delay(1); // waits
for the servo to get there
}
```

Output:




MEGA